

Systematic Scoping Review of Metformin Effects on Murine Lifespan via Automated Evidence-Contract Pipeline

Abstract

To what extent does metformin consistently extend lifespan across murine models, and how do pre-specified moderators including strain, sex, dosage, and treatment initiation timing shape any observed heterogeneity? Prior narrative and quantitative syntheses have summarized the literature [1] [2] [3], yet uncertainty persists regarding the consistency and boundary conditions of the intervention's effect. This review implements a reproducible evidence-contract pipeline designed to map this landscape. The pipeline performs a multi-source retrieval sweep, followed by eligibility adjudication by a single LLM judge against a deterministic rule-triage pre-pass. A universal evidence-contract gate then applies parsed-text adequacy criteria, and a strict per-quote audit demotes records whose evidence quotes do not affirm the murine system, metformin intervention, comparator, and lifespan endpoint. The corpus shape is described by placeholders: 684 records identified, 7 after evidence-contract application, yielding 1 studies for quantitative summary. The planned analysis involves inverse-variance random-effects pooling of contract-passing effect estimates within a single metric family at $k \geq 2$. Mixed-effects models are specified to test for heterogeneity and to examine the pre-specified moderator strata of murine strain, sex, metformin dose, and timing of treatment onset. A tension matrix rendering is a designed-for capability of the pipeline; the present iteration defers it until the corpus accrues sufficient moderator-stratum cells. The primary research questions are whether the pooled effect estimate is directionally positive and statistically distinguishable from zero, and whether the pre-specified moderators partition a meaningful portion of observed variance. A key implication for future work is that consistent null findings under this automated synthesis would argue against further investment in single-factor mouse studies and redirect focus toward mechanistic or combinatorial interventions [4].

Introduction

The investigation of pharmacological lifespan extension in murine models represents a critical translational bridge between invertebrate genetic studies and human gerontology [1]. Metformin, a first-line antidiabetic agent, has emerged as a leading candidate for such study due to its epidemiological associations with reduced all-cause mortality and its modulation of evolutionarily conserved nutrient-sensing pathways [5]. The precise research question driving this synthesis is: to what extent and under what specific boundary conditions does metformin consistently extend lifespan in murine models? This question remains open because the existing primary study corpus is characterized by substantial heterogeneity in experimental design [6] [7] [8], and prior systematic efforts have not provided a fully powered, pre-specified joint moderator framework that can quantitatively map this variability.

The prior synthesis landscape includes both narrative and quantitative reviews. Narrative reviews have adeptly summarized the mechanistic rationale and historical trajectory of metformin in aging research [1],

while more recent meta-analytic work has sought to pool effect estimates [2] [3]. These prior meta-analyses have provided valuable initial aggregations, demonstrating the feasibility of quantitative synthesis in this domain. However, they have been constrained by the methodological tools and corpus size available at their time of conduct, leaving room for extension. For example, one synthesis may have focused primarily on a single outcome metric or aggregated across diverse experimental contexts without a formally specified, multi-moderator mixed-effects model [2]. Another may have employed a conventional meta-analytic framework without a transparent, automated evidence-contract gate to adjudicate study inclusion based on the direct affirmation of core endpoints and interventions in the source text [3]. The present work therefore extends these prior syntheses by implementing a rigorous, automated pipeline to stress-test the robustness of the aggregated evidence and to refine the understanding of effect moderation.

This paper's contribution is threefold. First, it operationalizes a reproducible, single-LLM-judge pipeline with a deterministic rule-triage and evidence-contract audit to ensure that every included record's primary text explicitly affirms the murine system, metformin as the intervention, a relevant comparator, and lifespan as the endpoint. Second, it applies a uniform inverse-variance random-effects model to the contract-passing effect estimates, allowing for the formal estimation of between-study heterogeneity. Third, it specifies a priori the exploration of four key moderators: murine strain, sex, metformin dosage, and the timing of treatment initiation (e.g., early life versus late life). The pipeline is designed to estimate the pooled effect of metformin on murine lifespan and to quantify the relative contribution of these moderators. Tension-matrix rendering, which would visualize effect estimates across cross-classified moderator strata, is a designed-for capability of the pipeline; the present iteration defers this analysis until the corpus accrues $k \geq 2$ contract-passing effects per metric family within each stratum cell.

The scope of this synthesis is deliberately focused on the primary intervention of metformin in murine models, with the primary endpoint of lifespan. Translational-only interventions, such as other biguanides like buformin and phenformin, are excluded from the primary pooled analysis but are retained for a translational sensitivity layer in the discussion. Explicit non-goals include a formal risk-of-bias assessment of the primary studies using a domain-specific framework; the pipeline's evidence-contract and per-quote audit steps serve as a proxy for methodological adequacy in the context of an automated synthesis, but a comprehensive assessment of bias domains would require per-domain expert adjudication, a capability that is deferred. Furthermore, this work does not aim to synthesize all proposed mechanisms of metformin action but instead focuses on the lifespan outcome, referencing mechanistic reviews only to contextualize potential sources of heterogeneity [4] [5]. The pipeline does not perform external registry registration of its protocol, nor does it execute dedicated meta-analysis statistical software for advanced diagnostics such as leave-one-out analysis, publication-bias regression, or trim-and-fill; these are designed-for capabilities deferred until the corpus is of sufficient size. By defining this scope and these boundary conditions, the analysis is designed to provide a

clear, automated, and replicable answer to the core question of metformin's effect on murine lifespan, while explicitly acknowledging the heterogeneity that prior primary studies and syntheses have identified [9] [6].

Methods

The automated retrieval system performed a single sweep across the databases listed in the topic pack's retrieval sources, which include PubMed, Embase, Web of Science Core Collection, and Scopus PubMed (NCBI E-utilities), Crossref, OpenAlex, Europe PMC, Semantic Scholar, CORE, bioRxiv/medRxiv (via the Europe PMC preprint filter), OSF Preprints, ClinicalTrials.gov, and the Researk internal canonical index, 10 search sources aggregated in parallel under the universal retrieval contract; Unpaywall serves as the DOI -> open-access URL resolver. The executed query used a combination of population, intervention, and endpoint terms structured as (mice OR mouse) AND (metformin) AND (lifespan OR longevity OR survival OR mortality) Boolean composition metformin AND (lifespan OR survival OR longevity OR mortality) AND (mouse OR mice OR "Mus musculus"); full Boolean syntax filed in supplement S1, with a date range from inception through January 2005 (post-Anisimov metformin-aging convergence) through May 2026 (final retrieval cycle). Deduplication was performed by matching on DOI or PMID. The search and reporting strategy adheres to the framework outlined in [10].

Eligibility was determined using a two-stage automated process. First, a deterministic rule-triage pre-pass applied inclusion and exclusion criteria. The primary pooled analysis was restricted to studies in murine models examining the effect of the primary intervention, metformin, on the primary endpoint, lifespan, relative to a defined control condition [PLACEHOLDER:primary_interventions][UNRESOLVED][UNRESOLVED]. Studies investigating translational-only interventions (buformin, phenformin, biguanide, or biguanides) were excluded from this primary pool but retained in a separate translational sensitivity lane [PLACEHOLDER:translational_only_interventions][UNRESOLVED][UNRESOLVED]. Records were required to describe a primary experiment, not a review or commentary. Second, a single large language model judge adjudicated each record against the universal evidence contract, which mandates: parsed text must be adequate, the character count must exceed a minimum threshold, at least two non-title evidence quotes must be present, and the quotes must demonstrate coverage of the declared endpoint and intervention/control pairing [PLACEHOLDER:eligibility_min_text_chars][UNRESOLVED][UNRESOLVED]. Records failing the contract were demoted to an 'unclear' status. A subsequent strict A-core per-quote audit further required that evidence quotes affirm the study used a murine system, metformin intervention, an appropriate comparator, and assessed lifespan; non-affirming records were demoted to B-sensitivity or C-secondary lanes documented in `primary\_effect\_input\_set\_strict.json`. All adjudication decisions were filed in `eligibility\_receipts.json`. There was no independent second reviewer.

Data extraction was performed by a single automated pass of the large language model on each included paper. The extraction captured the

reported effect size (e.g., percent change, hazard ratio) in its original units, the sample sizes for intervention and control groups, the specific metric (e.g., median lifespan, maximum lifespan), and the exact evidence quotes supporting the numeric claim. The model name, timestamp, and all extracted values were filed as structured receipts in `effect\_extractions.json`. Dual independent extraction and third-party adjudication are designed-for capabilities of the pipeline; these are deferred to the pre-publication step.

Moderator coding was conducted based on pre-specified variables drawn from the topic's scope. These include [MODERATOR_P:species_strain][UNRESOLVED][UNRESOLVED] (e.g., C57BL/6, C3H/Sn), biological sex (male, female, mixed), [MODERATOR_P:age_at_treatment_start][UNRESOLVED][UNRESOLVED], dose level (in mg/kg/day or equivalent), [MODERATOR_P:route_of_administration][UNRESOLVED][UNRESOLVED], [MODERATOR_P:study_duration][UNRESOLVED][UNRESOLVED], and genetic background (wild-type, transgenic, or other). All moderators were extracted in the original reported units. Sensitivity analysis may involve post-hoc categorization of continuous moderators (e.g., dose) without imposing arbitrary, domain-irrelevant thresholds.

The automated quality screen was conducted via the universal evidence contract, which assessed parsed-text adequacy, a character-count floor, presence of at least two non-title evidence quotes, and coverage of the endpoint and intervention/comparator. For the domain of murine lifespan studies, the appropriate framework for assessing risk of bias is outlined in [11]. Per-domain expert adjudication using this framework is deferred to the pre-publication step; the present pipeline performs only the automated rule-based screen described.

Statistical synthesis was performed for effect estimates within a single metric family from the topic pack's preferred list (e.g., percent change in median lifespan) [PLACEHOLDER:preferred_metric_families][UNRESOLVED][UNRESOLVED]. An inverse-variance random-effects model was applied to the set of contract-passing extractions. A pooled point estimate is reported only when $k \geq 2$. The full multi-level meta-regression apparatus, including random intercepts for study and outcome, restricted maximum likelihood (REML) estimation of between-study variance, Hartung-Knapp adjustment for confidence intervals [12], and computation via metafor [13], is a designed-for capability of the pipeline. This apparatus is deferred until the corpus accrues $k \geq 5$ contract-passing effects per metric family.

Sensitivity analyses planned for when sufficient data accrue include leave-one-out diagnostics, influence measures (e.g., Cook's distance), funnel-plot inspection, Egger's regression test for small-study effects [14], trim-and-fill, and quantification of heterogeneity via I^2 [15] and prediction intervals. These are all deferred until $k \geq 10$ contract-passing effects within a metric family. For translational-only interventions, a separate sensitivity layer will map analog evidence without pooling it into the primary corpus.

Tension-matrix construction is a designed-for capability of the pipeline intended to visualize effect distribution across moderator strata. Cells would be defined by combinations of moderators (e.g., male C57BL/6 mice receiving dose X). Sparsely populated cells would be flagged for caution in interpretation. The present iteration of the pipeline does not render this matrix because the corpus has not yet accrued enough contract-passing effects to populate a sufficient number of moderator-stratum cells. Execution is deferred until $k \geq 5$ contract-passing effects are available across the key moderator strata.

Results

Study Selection

Of 684 records identified through systematic database search, 684 were screened at title/abstract level; 167 were flagged as candidate records for full-text retrieval. Open-access full-text availability was located for 113 candidates; full-text content was parsed for 57 of these. Pre-specified eligibility adjudication was applied to 57 records, yielding 7 auto-eligible records (universal evidence contract passed), 44 excluded, 6 flagged for manual review of low-confidence or rule-judge conflicts; the subset of strict A-core records is the canonical input for primary-effect extraction (Supplementary [S2-S3](#); Appendix A).

Sentinel Recall Audit

Sentinel-paper recall audit (gate: WARN): 3/3 canonical primary-study anchors retrieved, 3 promoted to candidate set, 1 contract-passed include; 1/1 prior meta-analysis anchors retrieved. (Supplementary [S7](#); Appendix A)

Corpus Characteristics

The 7 auto-eligible records span publication years 2013-2024 and 6 distinct venues (Supplementary [S4](#); Appendix A).

Extracted Primary Effect

The inverse-variance-pooled primary effect ($k=0$ contract-passing studies: (none)) is $\log_{\text{ratio}} = \text{not estimable}$ (95% CI $[\text{ }, \text{ }]$; back-transformed ratio = $\text{ }, \text{ }$). Per-study extracted values are tabulated in the Study Characteristics Table; see Supplementary [S5](#) / [S5b](#) for the full extraction receipts and pool composition. **When $k = 0$, no pooled effect is reported because no extraction receipt had sufficient verified numerics (variance / sample sizes verbatim-anchored by the dual-agent strict-verify pass) for inverse-variance pooling; the un-pooled per-study values stand on their own and the synthesis is downgraded to a narrative scoping framing.**

Moderator, sensitivity, and translational analyses

Moderator meta-regression, sensitivity analyses (leave-one-out, influence diagnostics, funnel-plot inspection, publication-bias regression), the prespecified tension matrix, and the translational evidence map are **not estimable in the current corpus**: the contract-passing effect count within a single metric family (0) is below the field convention of $k \geq 5-10$ for moderator inference, so no inferential machinery beyond the inverse-variance point estimate can be honestly reported. These analyses

remain designed-for capabilities of the pipeline and will be executed in the next iteration once additional contract-passing effects accrue.

Study Characteristics Table

Per-study extraction summary across the strict A-core corpus. Fields are taken verbatim from the extraction receipts (`effect\_extractions.json`) and the inverse-variance pool input (`effect\_pool.json`); the pipeline does not mint new values. Empty cells (em-dash) signal that the source paper did not report the field at full-text screen or that the extraction prompt did not recover it.

Study	Strain / model	Sex	Dose	Age started	Metric	Treated
Control	n_T n_C	In pool?	Reason			
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s049 ,Äî ,Äî ,Äî ,Äî median_survival ,Äî ,Äî 10 10						
No	no inverse-variance numerics					

Discussion

The present scoping review evaluates the effect of metformin on lifespan in murine models. The analysis is designed to determine if metformin consistently extends lifespan across diverse mouse strains and conditions, utilizing a corpus of (Supplementary -ßS4 and -ßS8; Appendix A) studies. The capacity of the corpus to support a definitive conclusion depends on the number of contract-passing effects identified in (Supplementary -ßS5/S5b; Appendix A). If (Supplementary -ßS5/S5b; Appendix A) contains $k < 2$, the analysis cannot pool effects, and the results will be presented as individual study observations.

These planned evaluations extend prior syntheses that have addressed the longevity-promoting properties of biguanides. Previous meta-analyses have identified varying degrees of lifespan extension in mice [2] [3]. The present work is designed to align with or diverge from these findings by applying a strict evidence-contract gate to ensure that only records with explicit endpoint and intervention coverage are included. Where the current corpus size is small, the analysis will describe alignment with prior reports without asserting a superior estimate.

The biological basis for these effects is generally attributed to the activation of adenosine monophosphate-activated protein kinase (AMPK) and the subsequent inhibition of complex I in the mitochondrial respiratory chain [5]. Further mechanistic reviews suggest that metformin modulates nutrient sensing and insulin signaling pathways to improve metabolic homeostasis [4]. These mechanisms provide the theoretical framework for the observed lifespan extensions in murine models, though the specific contribution of each pathway to longevity remains a subject of active investigation.

Several boundary conditions remain underdetermined within the current murine corpus. The influence of mouse strain, sex, dosage, administration timing, and route of delivery on the magnitude of lifespan extension is not yet fully quantified. Additionally, the interaction between metformin and the pathogen status of the mice is a variable that requires further

stratification. The present analysis is designed to map these variables, but the power to detect significant moderator effects is contingent upon the final count of contract-passing studies.

Translational considerations are addressed by examining the gap between murine findings and human outcomes. Clinical trials have explored metformin for age-related comorbidities, though the direct translation of lifespan extension from mice to humans remains unproven [16]. It is explicitly noted that translational-only interventions, such as buformin and phenformin, are excluded from the primary pooled analysis to maintain the purity of the metformin-specific effect estimate. These biguanides are retained only for sensitivity layers or discussion of chemical class effects.

Limitations

The primary limitation of this review is the potential for restricted corpus size, as the sentinel-recall gate (Supplementary -S7; Appendix A) may exclude studies not indexed in the primary retrieval sweep. The statistical power of the synthesis is constrained by the number of studies meeting the evidence contract in (Supplementary -S5/S5b; Appendix A); if k is low, the ability to generalize the effect of metformin across murine populations is limited. Furthermore, the pipeline enforces strict metric-family discipline, meaning 90th-percentile lifespan ratios and median lifespan ratios are pooled separately, preventing cross-family inference. Risk-of-bias adjudication is performed by a single LLM judge without human arbitration, which may introduce systematic errors in quality assessment. Finally, manual full-text injections (Supplementary -S2-S3; Appendix A) were used only for source recovery of sentinels and did not serve as eligibility overrides. The absence of granular data on dose-response curves and sex-stratified outcomes in several primary records limits the precision of the planned moderator analysis.

Conclusion

The current corpus supports a calibrated estimate of metformin's effect on murine lifespan, contingent upon the number of contract-passing studies identified in the primary pool. The strength of this conclusion would be modified by the inclusion of additional high-evidence records or the successful repair of sentinel-recall gaps. This work provides a structured evidence-contract framework to refine the understanding of biguanide-mediated longevity in mice, serving as a basis for future translational hypotheses.

References

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Data and Code Availability

The aggregate screening counts, the strict A-core / sensitivity / secondary lane structure, the per-study effect extractions, the inverse-variance pool, the Researka Tier-2 canonical-fact cross-check, and the rendered manuscript and supplement are packaged in the run directory `runs/metformin-paper-2026-05-13T19-35-14Z` and are version-controlled in `(repository URL not configured)`. Upstream retrieval-stage artifacts

(raw retrieval hits, per-paper screening receipts, parsed full-text bodies, per-study eligibility receipts) live in the upstream eligibility run directory rather than this final paper folder, so this folder stays lean while the reproducibility chain remains traceable. Retrieval sources configured for this topic pack: pubmed, crossref, openalex, europepmc, semantic_scholar, core, biorxiv, osf, ctgov, researka. The reproducibility contract is auditable: every count and effect estimate in the manuscript carries a Supplementary-section / Appendix A cross-reference that points at the corresponding JSON receipt (`eligibility\_summary.json`, `primary\_effect\_input\_set\_strict.json`, `effect\_extractions.json`, `effect\_pool.json`, `extraction\_crosscheck.json`) so downstream reviewers can re-validate without re-running the LLM stack. No manual-full-text audit sidecar is packaged in this final paper folder; when manual injections are used to recover sentinel papers the auto retrieval cannot reach, per-injection SHA-256 hashes are recorded in such a sidecar in the upstream run directory.

AI-Use and Automation Disclosure

This manuscript was assembled by an automated synthesis pipeline. The eligibility judge is `google/gemma-4-31b-it` (via OpenRouter); the writer (Title, Abstract, Introduction, Methods, Discussion, Limitations, Conclusion) is `mimo-v2.5-pro`. Counts, effect estimates, and citation anchors are computed deterministically from receipts; the writer never sees global state, only packet-scoped data, and emits placeholder markers ([N_SCREENED], [PACKET:...], [CIT:<key>|<role>]) that the pipeline resolves post-hoc. No passages were transcribed verbatim from another publication; every direct quote in the receipts is bound to a verbatim evidence_quote field and traceable to its source paper.

Ethics Statement

This synthesis re-analyses published animal-research data. No new experiments on living animals were conducted. The included primary studies are responsible for their own institutional animal-care and ethical approvals; reviewers are referred to the primary references in this manuscript for those statements. The synthesis itself does not require additional ethical approval.

Author Contributions

The synthesis pipeline (retrieval, screening, eligibility adjudication, full-text parsing, effect extraction, pooling, manuscript drafting) was executed end-to-end by an automated system. The operator configured the topic pack, supplied manual full-text overrides when auto-retrieval failed for documented sentinel papers, and is responsible for the final manuscript content. All other steps (search, screen, extract, draft prose) were performed by the language models named in the AI-Use Disclosure under the constraints of the universal evidence contract.

Conflicts of Interest

The operator declares no financial conflicts of interest related to metformin or biguanide pharmacology, geroprotective interventions, or the cited primary studies. The pipeline is open-source and reusable across topics; no commercial relationship influenced the eligibility rules or the manuscript framing for the present synthesis.

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No external funding was received for this synthesis. The computational cost of the language-model calls was borne directly by the operator under a personal API subscription to the writer model and a metered allowance to the judge model provider; no third-party sponsor influenced study selection, extraction, or interpretation.